5.1 Introduction to Probability

1. Probability of an Events

$$P(A) = \frac{\text{Number of favorable outcomes (Outcomes in A)}}{\text{Total number of possible outcomes}}$$

*Note: This counting formula works when all outcomes are equally likely

Range of Value: $0 \leq P(A) \leq 1$

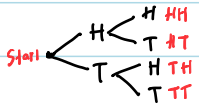
P(A)	Meaning
0	Impossible, it never happens
1	Certain, it always happen (100%)
0.3	30% Chance A happen

2. Sample Space and tree diagram

The sample space S is the set of all possible outcomes.

One coin: $S = \{H, T\}$

Two Coin: $S = \{HH, HT, TH, TT\}$



05.2 Probability Formulas, Symbols & Notations – Marginal, Joint, and Conditional Probabilities

1. Marginal probability – a single event

The probability of one event, regardless of any others:

$$P(A) = \frac{\text{Favorable Outcomes in event } A}{\text{Total Outcomes in Sample Space}}$$

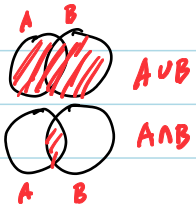
example 1: • Sample Space $S = \{1,2,3,4,5,6,7,8,9\}$

• "Event" $A = \{1,2,3,4\}$ $B = \{6,7,8\}$ $C = \{1,7,8,9\}$

$P(A) = \frac{4}{9} = 44.44\%$ $P(B) = \frac{3}{9} = 33.33\%$. $P(C) = \frac{4}{9} = 44.4\%$

2. Union and Intersection Symbols

Symbols	Name	Meaning
$A \cup B$	Union	Everything in A or B (or both), listed once
$A \cap B$	Intersection	Only what is in both A and B



Example 1:

$A \cup B = \{1,2,3,4,5,6,7,8\}$

$A \cap B = \{\text{None}\}$

$B \cup C = \{1,6,7,8,9\}$

$B \cap C = \{7,8\}$

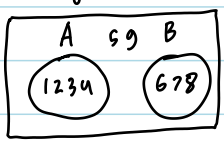
3. Union Probability – "A or B" Addition Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$P(A \cup B)$: $P(A$ or $B)$ — $P(A \cap B)$: $P(A$ and $B)$

The Subtraction stops $A \cap B$ being counted twice.

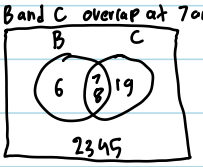
• Mutually exclusive events can't happen together, so $A \cap B \neq \emptyset$ and $P(A \cap B) = 0$: **Mutually Exclusive A, B**



$P(A \cup B) = \frac{7}{9} = 0.778 = P(A) + P(B) = \frac{4}{9} + \frac{3}{9}$

• Example overlap at 7 and 8: $B \cap C = \{7,8\}$ so $P(B \cap C) = \frac{2}{9}$: **Not Mutually Exclusive B, C**

B and C overlap at 7 and 8



$P(B \cup C) = \frac{5}{9} = P(B) + P(C) - P(B \cap C) = \frac{4}{9} + \frac{3}{9} - \frac{2}{9} = \frac{5}{9}$

4. Joint Probability - $P(A \cap B)$ (multiplication rule)

- $P(A \cap B)$ is read "The probability of A and B" (both happen):
  - ↳ Words: $P(A \text{ and } B) = P(A \text{ given } B)\ P(B) = P(B \text{ given } A)\ P(A)$
  - ↳ Symbols: $P(A \cap B) = P(A|B)\,P(B) = P(B|A)\,P(A)$

  On the **Independent events** (one doesn't affect the others): $P(A|B) = P(A)$ and $P(B|A) = P(B)$, so

  Words: $P(A \text{ and } B) = P(A)\,P(B)$

  Symbols: $P(A \cap B) = P(A)\,P(B)$

- Independent or dependent?
  - Situation: Draw a red marble, **Put it back**, then draw blue one

    Type: Independent.

    Why: Replacing the marble restore the bag.
  - Situation: Draw a red marble, **keep it**, then draw blue one

    Type: Dependent.

    Why: The bag now has one fewer marble, so the second probability change.
  - Situation: Toss a coin Twice

    Type: Independent

    Why: The first toss doesn't affect a second toss.

- Notation Warning: $P(A \cap B)$ vs $P(AB)$
  - $P(A \cap B) = P(B \cap A)$: both happen, no order implied.
  - Some books write $P(AB)$ for "A first, then B" and $P(BA)$ for "B first, then A". These are not always equal.
  - Read the context: are the events simultaneous, or does one come before the other?

5. Conditional Probability - "A given B"

$P(A|B)$ is the probability of A **given that** B **has already happened**.

we know that $P(A \cap B) = P(A|B)\,P(B) = P(B|A)\,P(A)$

So

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \qquad P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Conditional = Joint / Marginal

6. Bayes' Theorem

Replacing $P(A \cap B)$ with $P(B|A)\,P(A)$

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

Replacing $P(A \cap B)$ with $P(A|B)\,P(B)$

$$P(B|A) = \frac{P(A|B)\,P(B)}{P(A)}$$

7. Negation (Complement) Probability

Words: P(not A) = 1 - P(A)

Symbols: $P(A') = 1 - P(A)$

$P(A')$ is read "The probability of Not A". The complement is also written A^c (The most common form in textbook), $\bar{A}$ or $\sim A$

example 1:

$P(A') = 1 - P(A) = 1 - \frac{4}{9} = \frac{5}{9}$

$P(B) = 0.35$ then $P(B') = 1 - 0.35 = 0.65$, a 65% chance that B doesnot occur.

>Formula Summary

Type	Words	Symbols
Marginal	$P(A) = \frac{\text{favorable}}{\text{total}}$	(same)
Union (or)	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Union, mutually exclusive	$P(A \text{ or } B) = P(A) + P(B)$	$P(A \cup B) = P(A) + P(B)$
Joint (and)	$P(A \text{ and } B) = P(A \text{ given } B)P(B) = P(B \text{ given } A)P(A)$	$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A)$
Joint, independent	$P(A \text{ and } B) = P(A)P(B)$	$P(A \cap B) = P(A)P(B)$
Conditional	$P(A \text{ given } B) = \frac{P(A \text{ and } B)}{P(B)}$	$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$
Bayes	$P(A \text{ given } B) = \frac{P(B \text{ given } A)P(A)}{P(B)}$	$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$
Complement	$P(\text{not } A) = 1 - P(A)$	$P(A') = 1 - P(A)$

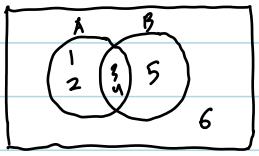
05.3 Probability of Complementary Events

Setup - rolling one six-sided die

- Sample Space $S = \{1,2,3,4,5,6\}$
- A = number at most 4 = $\{1,2,3,4\}$
- $B = \{3,4,5\}$

1. $P(A \text{ and } B)$ - Intersection - $P(A \cap B)$

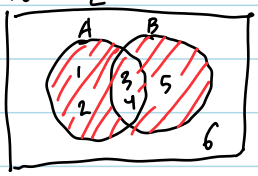
$A \cap B = \{3,4\}$ (the outcomes in both sets)



$$P(A \cap B) = \frac{2}{6} = \frac{1}{3} = 0.33$$

2. $P(A \text{ or } B)$ - Union - $P(A \cup B)$

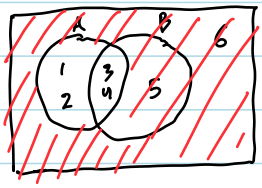
$A \cup B = \{1,2,3,4,5\}$ (the outcomes in either set; only 6 is missing)



$$P(A \cup B) = \frac{5}{6}$$

3. $P(\text{not } A)$ - Complement - $P(A')$

The complement A' is everything in S that is not in A: $A' = \{5,6\}$



4. $P(A)$ and the complement rule

$$P(A) = \frac{4}{6} = \frac{2}{3}$$

$$P(A) + P(A') = \frac{2}{3} + \frac{1}{3} = 1$$

↓ event ↓ complement of event A

An event and its complement together cover the whole sample space, so their probabilities add to 1:

$$P(A') = 1 - P(A)$$

05.4 Conditional Probability with Venn Diagrams and Contigency Tables

- **What it Means**

$P(A|B)$ is the probability that A happens given that B has already happen

Formula: $P(A|B) = \frac{P(A \cap B)}{P(B)}$ Word: $P(A \text{ given } B) = \frac{P(A \text{ and } B)}{P(B)}$

Shortcut: ask "how much of B is also A?" Once B is known to have happened,

$P(A|B) = \frac{\text{Count in both A and B}}{\text{Count in B}}$ → B becomes the new sample space.

- **Example 1 - a die**

$S = \{1,2,3,4,5,6\}$, $A = \text{odd} = \{1,3,5\}$, $B = \{3,4,5\}$

Shortcut: of the 3 outcomes in B, two (3 and 5) are in A:

$$P(A|B) = \frac{2}{3}$$

Formula:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{2/6}{3/6} = \frac{2}{3} \checkmark$$

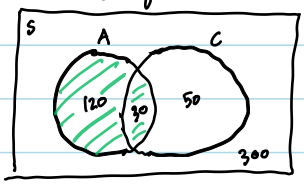
- **Example 2 - Venn diagram (algebra and Chemistry)**

A school has 500 students, 150 take algebra (A), 80 take chemistry (C), and 30 take both.

The four region of the venn diagram (two overlapping circles inside a box of 500):

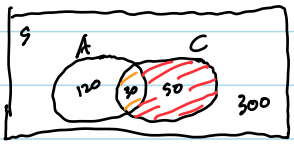
Region	Count
Algebra only (A but not C)	150 − 30 = 120
Both ($A \cap C$, the overlap)	30
Chemistry only (C but not A)	80 − 30 = 50
Neither (outside both circle)	500 − 200 = 300
Total	500

(a) $P(\text{taking Algebra}) = P(A)$



$P(A) = \frac{150}{500} = \frac{3}{10} = 0.3$

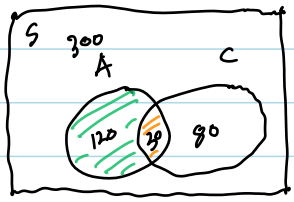
(b) $P(\text{Algebra} \mid \text{Chemistry}) = P(A|C)$



$P(A|C) = \frac{30}{80} = 0.35$

$= \frac{P(A \cap C)}{P(C)} = \frac{30/500}{80/500}$

(c) $P(\text{Chemistry} \mid \text{Algebra}) = P(C|A)$



$P(C|A) = \frac{P(C \cap A)}{P(A)}$

$= \frac{30/500}{150/500} = 0.2$

- Example 3 - Contingency Table (birds in a Zoo)

200 birds: - 70 male with brown eyes $|M \cap Br| = 70$ $n(M \cap Br)$
- 100 femal with brown eyes $|F \cap Br| = 100$ $n(F \cap Br)$
- 20 male with brown eyes $|M \cap Br| = 20$ $n(M \cap Br)$
- 10 female with blue eyes $|F \cap Bl| = 10$ $n(F \cap Br)$

* $|\cdot|$ = Cardinality (number of elements in a set)
* $n(\cdot)$ = Cardinality Function

> Birds

Eye color / Gender	Br Brown Eyes	Bl Blue Eyes	Total
Male	70	20	90
Female	100	10	110
Total	170	30	200

a) P(female)

$P(F) = \frac{110}{200}$

b) P(male and brown eyes)

$P(M \cap Br) = \frac{20}{200}$

c) P(female | brown eyes)

$P(F|Br) = \frac{100}{170}$ or $P(F|Br) = \frac{P(F \cap Br)}{P(Br)} = \frac{100/200}{170/200}$

d) P(blue eyes | Female)

$P(Bl|F) = \frac{10}{110}$

e) P(male | Blue eyes)

$P(M|Bl) = \frac{P(M \cap Bl)}{P(Bl)} = \frac{20/200}{30/200} = 0.67$

Joint vs conditional probability in a contingency table

Orange = numerator. Blue outline = denominator. Light blue = the row or column you restrict to.

(b) P(male and brown)

	Blue	Brown	Total
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

70 / 200 = 0.35

joint: divide by the grand total

(c) P(female | brown)

	Blue	Brown	Total
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

100 / 170 ≈ 0.588

given brown: divide by the column total

(e) P(blue | female)

	Blue	Brown	Total
Male	20	70	**90**
Female	10	100	**110**
Total	**30**	**170**	**200**

10 / 110 ≈ 0.091

given female: divide by the row total

The numerator is always one cell. What changes is the denominator: the grand total for a joint probability, or one row or column total for a conditional probability.

Key takeaways

- $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$, or with counts, $\dfrac{n(A \cap B)}{n(B)}$.
- The "given" event becomes the **denominator**, the new reduced sample space.
- In a contingency table, a conditional probability uses **one row or column total** as the denominator. A joint probability uses the **grand total**.
- $P(A \mid B)$ and $P(B \mid A)$ are usually different.

05.5 Probability of Independent and Dependent Events

1. Notation "A then B" as an Intersection

The video says "P(A then B)". Formally this is an **intersection** ("and"), with the order recorded by subscripts:

A_1 = "A happens on the 1st draw" $\qquad P(A \text{ then } B) = P(A_1 \cap B_2)$

B_2 = "B happens on the 2nd draw"

The symbol $\cap$ means "and". Since $A \cap B = B \cap A$, the order ("then") comes from the subscript, not from $\cap$

- **The Multiplication Rule (General Form)**

From the Definition of Conditional Probability $P(B|A) = \frac{P(A \cap B)}{P(A)}$, multiplying both sides by $P(A)$ gives:

$$P(A_1 \cap B_2) = P(A_1) \times P(B_2|A_1)$$

2. Definitions

- **Independent events:** The outcome of one does **not** change the probability of the other, i.e. $P(B_2|A_1) = P(B_2)$, so the condition drops out:

$$P(A_1 \cap B_2) = P(A_1) \cdot P(B_2)$$

- **Dependent events:** The first outcome **changes** the probability of the second, so the conditional probability must be kept:

$$P(A_1 \cap B_2) = P(A_1) \cdot P(B_2|A_1)$$

Drawing marbles...	Type	Why
with replacement	Independent	The marble goes back, so the bag is unchanged
without replacement	Dependent	The bag has one fewer marble (and maybe one fewer of a color)

- **Example ~ A bag of Marbles**

	Red	Blue	Green	Yellow	Total
n(·)	8	7	6	4	25

* Notation: R_1 = red on draw 1, G_2 = green on draw 2, and so on.

T_2 = The set of marbles available for the second draw.

(a) P(red)

$$P(R) = \frac{8}{25} = 0.32$$

(b) P(Blue, then Green, with replacement)

$$P(B_1 \cap G_2) = \frac{7}{25} \times \frac{6}{25} = \frac{42}{625} = 0.0672$$

(c) P(yellow, then red, without replacement)

$$P(Y_1 \cap R_2) = P(Y_1) \times P(R_2|Y_1) = \frac{n(Y_1)}{n(T_1)} \times \frac{n(R_2)}{n(T_2)} = \frac{4}{25} \times \frac{8}{24} = \frac{4}{75} = 0.0533$$

(d) Two Blue, with replacement

$$P(B_1 \cap B_2) = P(B_1) \cdot P(B_2) = \frac{n(B_1)}{n(T_1)} \times \frac{n(B_2)}{n(T_2)} = \frac{7}{25} \times \frac{7}{25} = 0.0784$$

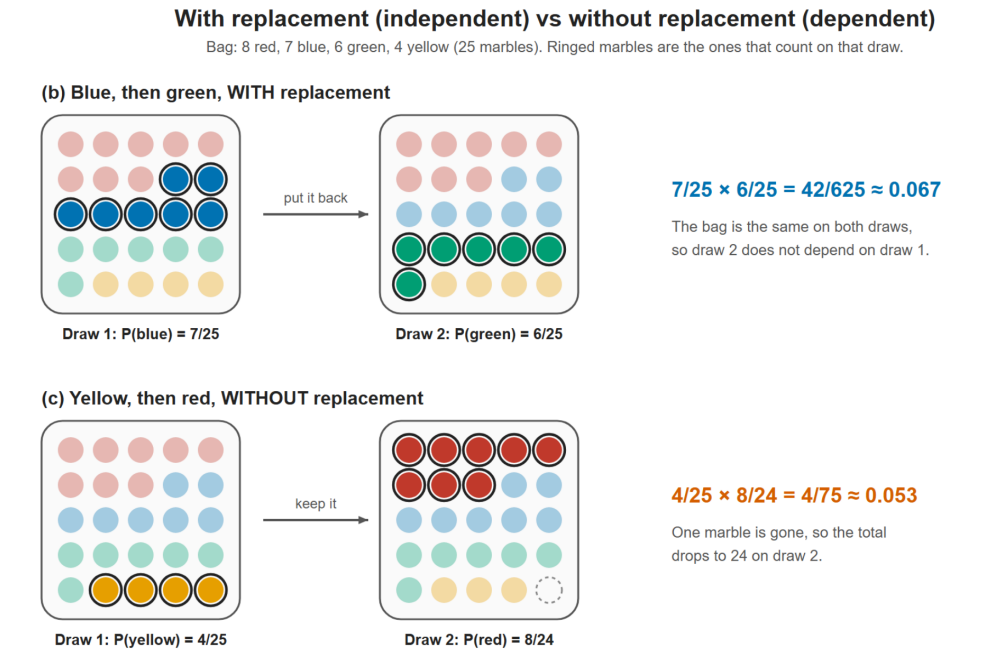
(e) Two Green, without replacement

$$P(G_1 \cap G_2) = P(G_1) \times P(G_2|G_1) = \frac{n(G_1)}{n(T_1)} \times \frac{n(G_2)}{n(T_2)} = \frac{6}{25} \cdot \frac{5}{24} = 0.05$$

After one green marble is removed, there are 5 greens out of 24

Why (b) is independent and (c) is dependent

- In (b), $P(G_2|B_1) = \frac{6}{25}$, which is exactly $P(G_2)$ on its own. The first draw has no effect.
- In (c), $P(R_2|Y_1) = 8/24$, instead of $P(R_2) = \frac{8}{25}$, the first draw changes it.



3. Key Takeaways

- With replacement → Independent: multiply the unchanged probabilities.
- Without replacement → Dependent: after each draw, reduce the total and (if needed) the count of the color drawn.
- "First... then..." problems use the multiplication rule: $P(A_1 \cap B_2) = P(A_1)\, P(B_2|A_1)$
- For three or more draws it chains: $P(A_1 \cap B_2 \cap C_3) = P(A_1) \cdot P(B_2|A_1) \cdot P(C_3|A_1 \cap B_2)$

5.6. Probability of Mutually Exclusive Events with Venn Diagrams

1. Definition

Events A and B are **mutually exclusive** if they **cannot happen at the same time**, meaning they share **no outcomes**:

$A \cap B = \emptyset \iff P(A \text{ and } B) = 0$

Venn Diagram: mutually exclusive events are drawn as **separate, non-overlapping** circles

events that are not mutually exclusive are drawn as **overlapping** circles.

2. Testing for Mutual Exclusivity

One die, $S = \{1,2,3,4,5,6\}$

- $A = \{1,2,3\}$, $B = \{5,6\}$, $C = \{3,4,5\}$

Rule: • If $P(A \cap B) = 0$, They are mutually exclusive.
If it is anything other than 0, They are not.

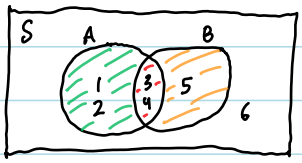
Pair	Intersection	P(both)	Mutually Exclusive?
A, B	$\emptyset$	0	yes
A, C	3	1/6	No
B, C	5	1/6	No

3. The Addition Rule

- **General (always works):** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- **Mutually Exclusive (The last term is 0):** $P(A \cup B) = P(A) + P(B)$

• Example 1 - Overlapping Events

$A = \{1,2,3,4\}$, $B = \{3,4,5\}$, $S = \{1,2,3,4,5,6\}$

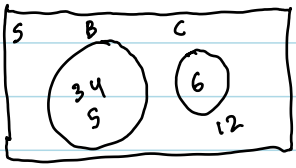


$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= \frac{4}{6} + \frac{3}{6} - \frac{2}{6} = \frac{5}{6}$$

• Example 2 - Mutually Exclusive Events

$B = \{3,4,5\}$ $C = \{6\}$

$$P(B \cup C) = P(B) + P(C)$$
$$= \frac{3}{6} + \frac{1}{6}$$
$$= 4/6$$



4. Key Takeaways

- Mutually exclusive = no shared outcomes = $P(A \cap B) = 0$
- use $P(A) + P(B) - P(A \cap B)$ in general. For mutually exclusive $P(A \cap B) = 0$ then it will be $P(A) + P(B)$
- A probability above 1 is a sign that an overlap was counted twice.

5.7 Addition Rule and Multiplication Rule

1. Addition Rule - "Or"

$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$

- for Mutually Exclusive events $P(A \cap B) = 0 \rightarrow P(A \cup B) = P(A) + P(B)$
- The general form Always works.

2. Multiplication Rule - "And"

Start from Conditional Probability:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \rightarrow P(A \cap B) = P(B) \times P(A|B)$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \rightarrow P(A \cap B) = P(A) \times P(B|A)$$

Independent events: $P(A|B) = P(A)$ and $P(B|A) = P(B)$, so

$P(A \cap B) = P(A) \times P(B)$

Note: "$A \cap B$" means both happen together, so $P(A \cap B) = P(B \cap A)$ this is not the same as "A then B" or "$A_1 \cap B_2$", where Order can Matter.

3. Test

Property	Test
Independent	$P(A\|B) = P(A)$, or equivalently $P(A \cap B) = P(B) \times P(A\|B) = P(A)P(B)$
Mutually Exclusive	$P(A \cap B) = 0$

4. Example - Sarah's Course Choice

- $P(A) = P(\text{enrolls in Algebra}) = 0.3$
- $P(B) = P(\text{enrolls in Biology}) = 0.7$
- $P(A|B) = P(\text{Algebra given Biology}) = 0.4$

* Note: we know that $P(A|B) = \frac{P(A \cap B)}{P(B)}$

(a) $P(\text{algebra and biology})$ - multiplication rule

$P(A \text{ and } B) = P(A \cap B) = P(B) \times P(A|B) = 0.7 \times 0.4 = 0.28$

(b) $P(\text{algebra or biology})$ - addition rule

$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.3 + 0.7 - 0.28 = 0.72$

(c) Are the events Independent?

$P(A) \neq P(A|B)$ → so the events are not independent: taking biology makes algebra more likely

(d) Are the events mutually exclusive?

$P(A \cap B) = 0.28 \neq 0$, so they are not mutually